

Lecture 1

Groups, examples, and the landscape of finite groups

Objectives. We abstract the group axioms from the behaviour of symmetry operations, assemble a working zoo of finite groups — cyclic, dihedral and symmetric — together with a first glimpse of the matrix groups of Part II, make precise when two groups are “the same”, and prove the fundamental counting theorem of the subject, Lagrange’s theorem.

1.1 From symmetry operations to the group axioms

Symmetry, in physics as in geometry, is always symmetry *under something*: a transformation that leaves the relevant structure unchanged. The structure may be a geometric figure, a crystal lattice, a molecule’s equilibrium configuration, or — as we shall see at the end of Part I — the Hamiltonian of a physical system. The first task of this course is to extract, from the behaviour of such transformations, a small set of axioms, and then to develop the mathematics of those axioms far enough that it starts returning concrete physical predictions: degeneracies of energy levels, patterns of vibrational spectra, selection rules.

We begin with the simplest nontrivial example. Let Δ be an equilateral triangle in the plane, its vertices labelled 1, 2, 3. A *symmetry operation* of Δ is a distance-preserving map of the plane that carries Δ onto itself. There are exactly six (Figure 1.1):

- the identity map e , which moves nothing;
- the rotations r and r^2 about the centroid, through $2\pi/3$ and $4\pi/3$ anticlockwise;
- the reflections $\sigma_1, \sigma_2, \sigma_3$ in the three axes joining a vertex to the midpoint of the opposite side.

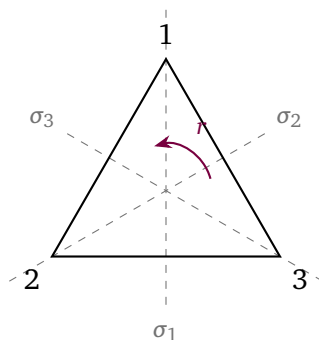


Figure 1.1: The six symmetry operations of the equilateral triangle: the identity, two rotations about the centroid, and reflections in the three dashed axes. The axis σ_i passes through vertex i .

Each operation is determined by what it does to the vertices. Writing $g: i \mapsto j$ when g carries vertex i to the position of vertex j , we have

$$r: 1 \mapsto 2, 2 \mapsto 3, 3 \mapsto 1, \quad \sigma_1: 1 \mapsto 1, 2 \mapsto 3, 3 \mapsto 2,$$

and similarly for the rest. Two operations performed in succession are again a symmetry operation. Throughout the course, composition is written *right to left*: gh means “first h , then g ”, so that $(gh)(x) = g(h(x))$ for points x . Tracing vertices,

$$r\sigma_1: 1 \xrightarrow{\sigma_1} 1 \xrightarrow{r} 2, \quad 2 \mapsto 3 \mapsto 1, \quad 3 \mapsto 2 \mapsto 3,$$

so $r\sigma_1$ fixes vertex 3 and swaps 1, 2: it equals σ_3 . The same computation in the other order gives $\sigma_1r = \sigma_2$. Hence

$$r\sigma_1 = \sigma_3 \neq \sigma_2 = \sigma_1r :$$

the order in which symmetry operations are performed *matters*. This noncommutativity is not a defect; it is the source of most of the interesting structure in this course.

Four features of this example are completely general. The composition of two symmetry operations is a symmetry operation (*closure*); composition of maps is associative; the identity map is a symmetry operation; and every symmetry operation can be undone by a symmetry operation (*inverses*). These are all that the abstract theory retains. The next definition lists three of them as axioms and builds the fourth, closure, into the requirement that the multiplication takes values in the set.

1.2 The definition of a group

Definition 1.1 (Group). A group is a set G together with a map $G \times G \rightarrow G$, $(g, h) \mapsto gh$, called the *group multiplication*, such that:

(G1) *Associativity*: $(gh)k = g(hk)$ for all $g, h, k \in G$.

(G2) *Identity*: there exists $e \in G$ with $eg = ge = g$ for all $g \in G$.

(G3) *Inverses*: for every $g \in G$ there exists $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

Remark 1.2 (Conventions). Closure is built into the definition: the multiplication is required to take values *in* G . We write the multiplication by juxtaposition and call gh the product, even when the operation is composition of maps or addition of numbers. The identity is denoted e throughout these notes; the physics and chemistry literature often writes E , and algebra texts sometimes write 1. By associativity, products such as ghk need no brackets, and powers $g^n = g \cdots g$ (n factors), $g^0 = e$, $g^{-n} = (g^{-1})^n$ obey the usual rules of exponents.

Definition 1.3 (Abelian group). A group G is *abelian* (or commutative) if $gh = hg$ for all $g, h \in G$. For abelian groups one sometimes uses additive notation: $g + h$ for the product, 0 for the identity, $-g$ for the inverse.

Example 1.4. (i) The integers $\mathbb{Z}$ under addition form an abelian group: the identity is 0 and the inverse of n is $-n$. Likewise $\mathbb{R}$ and $\mathbb{C}$ under addition. (ii) The nonzero rationals $\mathbb{Q} \setminus \{0\}$ under multiplication form an abelian group, as do $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$; the two-element set $\{1, -1\}$ is a group under multiplication. (iii) The natural numbers $\mathbb{N}$ under addition are *not* a group (no inverses), and $\mathbb{Z}$ under multiplication is not a group (2 has no integer inverse). (iv) The six symmetry operations of the triangle form a *nonabelian* group under composition, as we verified above.

Proposition 1.5 (Elementary consequences of the axioms). *Let G be a group. Then:*

- (i) *the identity element is unique;*
- (ii) *each $g \in G$ has a unique inverse;*
- (iii) *(cancellation) $gh = gk$ implies $h = k$, and $hg = kg$ implies $h = k$;*
- (iv) *$(gh)^{-1} = h^{-1}g^{-1}$ and $(g^{-1})^{-1} = g$.*

Proof. (i) If e and e' both satisfy (G2), then $e = ee' = e'$. (ii) If h and h' are both inverse to g , then $h = he = h(gh') = (hg)h' = eh' = h'$. (iii) Multiply $gh = gk$ on the left by g^{-1} and use associativity: $h = eh = (g^{-1}g)h = g^{-1}(gh) = g^{-1}(gk) = k$; right cancellation is analogous. (iv) $(h^{-1}g^{-1})(gh) = h^{-1}(g^{-1}g)h = h^{-1}h = e$, and similarly $(gh)(h^{-1}g^{-1}) = e$, so $h^{-1}g^{-1}$ is an inverse of gh and hence, by (ii), the inverse. Finally g inverts g^{-1} , so $(g^{-1})^{-1} = g$. ■

Note the order reversal in (iv): to undo “socks, then shoes” one removes the shoes first. For symmetry operations this is the obvious statement that undoing a composite transformation means undoing its steps in reverse order.

Definition 1.6 (Order). The *order* of a group G , written $|G|$, is the number of its elements; G is *finite* if $|G| < \infty$. The *order* of an element $g \in G$, written $\text{ord}(g)$, is the smallest integer $n \geq 1$ with $g^n = e$, if such an n exists; otherwise g has infinite order.

Proposition 1.7. *Let G be any group and let $g \in G$ have finite order $n = \text{ord}(g)$. Then for $m \in \mathbb{Z}$,*

$$g^m = e \iff n \mid m,$$

and the powers $e, g, g^2, \dots, g^{n-1}$ are distinct. If G is finite, then every element has finite order.

Proof. If $n \mid m$, say $m = qn$, then $g^m = (g^n)^q = e$. Conversely, if $g^m = e$, write $m = qn + s$ with $0 \leq s < n$ (division with remainder); then $e = g^m = (g^n)^q g^s = g^s$, and minimality of n forces $s = 0$. Distinctness follows: if $g^a = g^b$ with $0 \leq b < a < n$ then $g^{a-b} = e$ with $0 < a - b < n$, contradicting minimality.

For the last statement, let G be finite. The elements $g, g^2, g^3, \dots$ cannot all be distinct in a finite set, so $g^a = g^b$ for some $a > b \geq 1$; cancellation gives $g^{a-b} = e$ with $a - b \geq 1$, so g has finite order. ■

In the triangle group, the rotation r has order 3 and each reflection σ_i has order 2: a reflection is its own inverse.

1.3 A zoo of examples

Group theory is learnt from examples. The three families introduced in this section — cyclic, dihedral and symmetric groups — will be our laboratory for the whole of Part I.

1.3.1 Cyclic groups

Definition 1.8 (Cyclic group). A group G is *cyclic* if there is an element $g \in G$, called a *generator*, such that every element of G is a power of g :

$$G = \{g^k : k \in \mathbb{Z}\}.$$

The cyclic group with n elements is denoted C_n .

If a generator g has order n , then by Proposition 1.7 the powers $e, g, g^2, \dots, g^{n-1}$ are distinct and exhaust the group, and $g^a g^b = g^{a+b}$ with the exponent read modulo n . The abstract group C_n appears in many concrete guises:

- (a) the rotations of the plane through multiples of $2\pi/n$ about a fixed point, under composition;
- (b) the n -th roots of unity $\{1, \omega, \omega^2, \dots, \omega^{n-1}\} \subset \mathbb{C}$, where $\omega = e^{2\pi i/n}$, under multiplication;
- (c) the residues $\{0, 1, \dots, n-1\}$ under addition modulo n , written $\mathbb{Z}_n$.

That these are “the same group” is made precise in Section 1.3.4. Every cyclic group is abelian, since $g^a g^b = g^{a+b} = g^b g^a$. The integers $\mathbb{Z}$ under addition are the *infinite* cyclic group: every element is a multiple of the generator 1.

1.3.2 Dihedral groups

Definition 1.9 (Dihedral group). For $n \geq 3$, the *dihedral group* D_n is the group of symmetry operations of a regular n -gon, that is, of all distance-preserving maps of the plane carrying the n -gon onto itself, under composition.

(Beware: some books write D_{2n} for this group, labelling it by its order rather than by the polygon. We always label by the polygon.)

Proposition 1.10 (Structure of D_n). Let r denote the rotation through $2\pi/n$ about the centre of the n -gon, and let s be the reflection in a fixed symmetry axis. Then $|D_n| = 2n$, and

$$D_n = \{e, r, r^2, \dots, r^{n-1}, s, rs, r^2s, \dots, r^{n-1}s\},$$

where the first n elements are the rotations and the last n are the reflections. The generators obey

$$r^n = s^2 = e, \quad srs^{-1} = r^{-1}, \quad (1.1)$$

and consequently $sr^k = r^{-k}s$ for all k and

$$(r^a s^\varepsilon)(r^b s^\delta) = r^{a+(-1)^\varepsilon b} s^{\varepsilon+\delta}, \quad \varepsilon, \delta \in \{0, 1\}, \quad (1.2)$$

with the first exponent read modulo n and the second modulo 2.

Proof. A symmetry operation of the polygon permutes the vertices and so preserves the centroid. We use one standard fact of Euclidean geometry: a distance-preserving map of the plane that fixes a point is a linear orthogonal map about that point, hence a rotation or a reflection. Placing the centroid at the origin, the operation is therefore a rotation or a reflection. A rotation maps the vertex set to itself precisely when its angle is a multiple of $2\pi/n$, giving the n rotations $e, r, \dots, r^{n-1}$. A reflection does so precisely when its axis passes through two opposite vertices, two opposite edge midpoints, or a vertex and an opposite edge midpoint; counting in the even and odd cases separately, in both cases there are exactly n admissible axes, giving n reflections. Hence $|D_n| = 2n$.

If σ is any reflection, then σs preserves orientation, so it is a rotation: $\sigma s = r^k$ for some k , whence $\sigma = r^k s$ using $s^2 = e$. This exhibits the listed $2n$ elements as exhausting D_n , with the reflections being exactly the $r^k s$.

For the conjugation relation, compute the effect of srs^{-1} on the plane: $s^{-1} = s$ reflects, r rotates by $2\pi/n$ anticlockwise, and s reflects back; a rotation viewed in a mirror is a rotation

through the same angle in the opposite sense, so $srs^{-1} = r^{-1}$. (Sceptics may verify this by tracing two adjacent vertices, or by multiplying 2×2 orthogonal matrices.) Iterating gives $sr^k s^{-1} = r^{-k}$, i.e. $sr^k = r^{-k}s$, and (1.2) follows by moving all s 's to the right: when $\varepsilon = 1$, pushing s past r^b flips the sign of b . ■

The rule (1.2) makes computation in D_n entirely mechanical. For the triangle, D_3 is precisely the six-element group of Section 1.1, with $\sigma_1 = s$ (say), $\sigma_2 = r^2s$ and $\sigma_3 = rs$ (consistent with $r\sigma_1 = \sigma_3$ and $\sigma_1r = \sigma_2$ derived there).

Example 1.11 (Cayley table of D_3). Listing the elements in the order e, r, r^2, s, rs, r^2s and writing the product gh in row g and column h , rule (1.2) yields the *Cayley table* (multiplication table)

gh	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

Two features deserve comment. First, each row and each column contains every element exactly once (a *Latin square*); this holds in any finite group and is exactly the cancellation law, Proposition 1.5(iii): the map $h \mapsto gh$ is injective, hence bijective on a finite set. Second, the table is not symmetric about its diagonal — D_3 is nonabelian, as we already knew.

Remark 1.12 (Generators and relations). Proposition 1.10 says that every element of D_n is a word in r and s , and that every product of such words can be reduced using only the relations (1.1). One summarises this by the *presentation*

$$D_n = \langle r, s \mid r^n = s^2 = e, srs^{-1} = r^{-1} \rangle.$$

Presentations are a compact and convenient bookkeeping device, and we use them freely for the concrete groups of this course; the general theory of groups defined by arbitrary presentations is subtle (even deciding whether two words represent the same element can be algorithmically undecidable) and is not needed here.

1.3.3 Symmetric groups

Definition 1.13 (Symmetric group). The *symmetric group* S_n is the set of all bijections (permutations) $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, under composition of maps.

Proposition 1.14. S_n is a group, and $|S_n| = n!$.

Proof. Composites and inverses of bijections are bijections, composition of maps is associative, and the identity map is a bijection; so S_n is a group. A permutation is specified by choosing $\sigma(1)$ (n ways), then $\sigma(2)$ ($n - 1$ remaining ways), and so on: $n!$ permutations in all. ■

A permutation can be recorded in *two-line notation*, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ meaning $\sigma(1) = 3$ and so on, but *cycle notation* is far more useful. The symbol $(a_1 a_2 \dots a_k)$ denotes the k -cycle sending $a_1 \mapsto a_2 \mapsto \dots \mapsto a_k \mapsto a_1$ and fixing every other point. Cycles on disjoint sets of points commute. A 2-cycle $(a b)$, which merely swaps a and b , is called a *transposition*.

Proposition 1.15 (Disjoint cycle decomposition). Every permutation $\sigma \in S_n$ is a product of cycles on pairwise disjoint sets of points, and this decomposition is unique up to the order of the factors.

Proof. Follow the point 1 under iteration: $1, \sigma(1), \sigma^2(1), \dots$ must eventually repeat; injectivity of σ forces the first repetition to be 1 itself, closing a cycle $(1 \ \sigma(1) \ \dots \ \sigma^{k-1}(1))$. Now repeat from the smallest point not yet visited. The cycles so produced are disjoint by construction and their product is σ , since each point is moved by exactly one of them, in the same way as by σ . Uniqueness: the cycle containing a given point a is forced — it must read $(a \ \sigma(a) \ \sigma^2(a) \ \dots)$. ■

Example 1.16. In S_3 , with composition right to left,

$$(1 \ 2)(2 \ 3) = (1 \ 2 \ 3), \quad (2 \ 3)(1 \ 2) = (1 \ 3 \ 2),$$

as one checks by tracing each point; e.g. in the first product $1 \mapsto 1 \mapsto 2, 2 \mapsto 3 \mapsto 3, 3 \mapsto 2 \mapsto 1$. So S_3 is nonabelian — and indeed the computation should look familiar: it is the triangle computation of Section 1.1 in a new notation (there a rotation and a reflection refused to commute; here the two reflections $(1 \ 2)$ and $(2 \ 3)$ compose, in the two orders, to the two different rotations). We return to this in Example 1.22.

Proposition 1.17. Every permutation is a product of transpositions.

Proof. By Proposition 1.15 it suffices to decompose a cycle, and

$$(a_1 \ a_2 \ \dots \ a_k) = (a_1 \ a_k)(a_1 \ a_{k-1}) \cdots (a_1 \ a_2),$$

as one verifies by tracing each a_i through the right-hand side. ■

The decomposition into transpositions is far from unique — e.g. $e = (1 \ 2)(1 \ 2)$ uses two transpositions and e itself uses none — but the *parity* of the number of factors turns out to be an invariant. The clean way to see this is to let permutations act on a polynomial.

Definition 1.18 (Sign of a permutation). Let $\Delta(x_1, \dots, x_n) = \prod_{i < j} (x_i - x_j)$, and for $\sigma \in S_n$ let P_σ act on polynomials in $x_1, \dots, x_n$ by relabelling the variables:

$$(P_\sigma f)(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

Since P_σ permutes the factors of Δ up to signs, $P_\sigma \Delta = \pm \Delta$; the *sign* $\text{sgn}(\sigma) \in \{\pm 1\}$ is defined by $P_\sigma \Delta = \text{sgn}(\sigma) \Delta$.

This direct coordinate substitution is chosen to match our right-to-left composition convention, $(\sigma\tau)(i) = \sigma(\tau(i))$, and the next proposition checks that it does. (Lecture 3 will place the construction in its general setting, where the same substitution reappears for functions on any set the group acts on. Nothing in this lecture depends on that, and the remark can be passed over here.)

Proposition 1.19. $P_{\sigma\tau} = P_\sigma P_\tau$, and consequently $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma) \text{sgn}(\tau)$. Moreover every transposition has sign -1 , so if σ is a product of k transpositions then $\text{sgn}(\sigma) = (-1)^k$: the parity of k is independent of the decomposition.

Proof. It is enough to check an arbitrary monomial $x^\alpha = \prod_i x_i^{\alpha_i}$. Directly,

$$(P_\sigma x^\alpha)(x) = \prod_i x_{\sigma(i)}^{\alpha_i},$$

and hence

$$(P_\sigma P_\tau x^\alpha)(x) = \prod_i x_{\sigma(\tau(i))}^{\alpha_i} = \prod_i x_{(\sigma\tau)(i)}^{\alpha_i} = (P_{\sigma\tau} x^\alpha)(x).$$

Linearity gives $P_\sigma P_\tau = P_{\sigma\tau}$ for every polynomial. Applying this to Δ : $\text{sgn}(\sigma\tau)\Delta = P_\sigma(\text{sgn}(\tau)\Delta) = \text{sgn}(\sigma)\text{sgn}(\tau)\Delta$.

For the transposition $(a\ b)$ with $a < b$: the factor $(x_a - x_b)$ changes sign; factors involving neither a nor b are untouched; and the remaining factors, those involving exactly one of a, b , pair up as $(x_a - x_c)(x_b - x_c)$ with $c \neq a, b$, each pair being mapped to itself up to two compensating signs or to itself outright. Hence $P_{(a\ b)}\Delta = -\Delta$. The final statement follows from multiplicativity. ■

Permutations of sign $+1$ are called *even*, those of sign -1 *odd*. A k -cycle is a product of $k - 1$ transpositions (Proposition 1.17), hence has sign $(-1)^{k-1}$. The sign of a permutation will reappear throughout the course — most immediately in Lecture 3, where sgn is our first example of a nontrivial one-dimensional representation, and in Part II in the guise of the Levi-Civita symbol.

1.3.4 Comparing groups: isomorphism

The rotations through multiples of $2\pi/3$, the cube roots of unity, and the residues $\{0, 1, 2\}$ modulo 3 are visibly “the same group” in three different costumes: their elements match up so that the multiplication tables coincide. The following definition makes this precise.

Definition 1.20 (Isomorphism). An *isomorphism* between groups G and H is a bijection $\varphi: G \rightarrow H$ that preserves multiplication:

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2) \quad \text{for all } g_1, g_2 \in G.$$

If an isomorphism exists, G and H are *isomorphic*, written $G \cong H$.

An isomorphism automatically matches identities with identities and inverses with inverses: from $\varphi(e)\varphi(e) = \varphi(e)$ and cancellation in H , $\varphi(e) = e_H$; then $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$ gives $\varphi(g^{-1}) = \varphi(g)^{-1}$.

Example 1.21. The three realisations of C_n in Section 1.3.1 are isomorphic: writing ρ_k for rotation through $2\pi k/n$ and $\omega = e^{2\pi i/n}$, the maps

$$k \mapsto \rho_k \mapsto \omega^k$$

are bijections $\mathbb{Z}_n \rightarrow \{\rho_k\} \rightarrow \{\omega^k\}$ converting addition modulo n into composition of rotations into multiplication of complex numbers — in each case because $\rho_a \rho_b = \rho_{a+b}$ and $\omega^a \omega^b = \omega^{a+b}$, exponents modulo n . More generally, any two cyclic groups of the same order n are isomorphic via $g^k \mapsto h^k$: there is essentially *one* cyclic group of each order, which is why the notation C_n mentions only n .

Example 1.22 ($D_3 \cong S_3$). Every symmetry operation of the triangle permutes the vertex set $\{1, 2, 3\}$, so there is a map $\varphi: D_3 \rightarrow S_3$ sending each operation to the vertex permutation it induces. Performing two operations in succession permutes labels accordingly, and our composition conventions for maps of the plane and for permutations agree (rightmost factor first), so $\varphi(gh) = \varphi(g)\varphi(h)$. The map is injective: a distance-preserving map of the plane fixing three non-collinear points is the identity, so only e induces the trivial

permutation — and, combined with cancellation (the identities-and-inverses argument following Definition 1.20 used only $\varphi(gh) = \varphi(g)\varphi(h)$, so it applies to φ), if $\varphi(g) = \varphi(h)$ then $\varphi(h^{-1}g)$ is trivial, whence $g = h$. Since $|D_3| = 6 = 3! = |S_3|$, the injection φ is a bijection: an isomorphism. Explicitly, $r \mapsto (1\ 2\ 3)$ and $\sigma_1 = s \mapsto (2\ 3)$. For $n \geq 4$ the same map embeds D_n into S_n but is no longer surjective: $2n < n!$, and indeed not every permutation of the vertices of a square is realised by a rigid motion.

Remark 1.23 (Invariants, and the classification problem). Isomorphic groups share every property expressible in terms of the group multiplication alone: the order $|G|$, abelianness, and the multiset of orders of elements, among others. Such *invariants* are the practical tool for proving two groups are *not* isomorphic: on Sheet 1 you will check that D_4 and C_8 , both of order 8, are distinguished by their element orders. Classifying all finite groups up to isomorphism is the great organising problem of the subject; we survey what is known in Section 1.8.

1.4 Groups beyond the finite

Part I of this course concerns finite groups, but the matrix groups central to Part II cost nothing to define now, and they sharpen the picture of where the finite groups sit.

Definition 1.24 (Matrix groups). Let $\mathbb{F}$ denote $\mathbb{R}$ or $\mathbb{C}$. The *general linear group* $GL(n, \mathbb{F})$ is the set of invertible $n \times n$ matrices over $\mathbb{F}$ under matrix multiplication. Inside it sit, among others,

$$\begin{aligned} O(n) &= \{A \in GL(n, \mathbb{R}) : A^T A = \mathbb{1}\}, & SO(n) &= \{A \in O(n) : \det A = 1\}, \\ U(n) &= \{A \in GL(n, \mathbb{C}) : A^\dagger A = \mathbb{1}\}, & SU(n) &= \{A \in U(n) : \det A = 1\}, \end{aligned}$$

the *orthogonal*, *special orthogonal*, *unitary* and *special unitary* groups.

Example 1.25 ($O(n)$ is a group). Matrix multiplication is associative and $\mathbb{1}^T \mathbb{1} = \mathbb{1}$, so it suffices to check closure and inverses inside $GL(n, \mathbb{R})$. If $A^T A = B^T B = \mathbb{1}$ then $(AB)^T (AB) = B^T A^T AB = B^T B = \mathbb{1}$, and from $A^T A = \mathbb{1}$ we get $A^{-1} = A^T$ with $(A^{-1})^T A^{-1} = AA^T = AA^{-1} = \mathbb{1}$ (the relation $A^T A = \mathbb{1}$ for square matrices implies $AA^T = \mathbb{1}$). The same manipulations with $\dagger$ in place of T show $U(n)$ is a group, and the determinant conditions survive products and inverses since $\det(AB) = \det A \det B$.

These groups are *continuous*: their elements come in smooth families. For instance every element of $SO(2)$ is a rotation matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad R(\theta)R(\theta') = R(\theta + \theta'),$$

parametrised by an angle $\theta \in [0, 2\pi)$, and $O(2)$ consists of these rotations together with the reflections in all lines through the origin. Such groups — *Lie groups* — are the subject of Part II, where the ability to differentiate with respect to parameters like θ makes an entirely new range of methods available.

The finite groups we have met embed into this continuous landscape: the rotations of the regular n -gon form a copy of C_n inside $SO(2)$, and all its symmetry operations form a copy of D_n inside $O(2)$. In fact these are the only possibilities in the plane:

Remark 1.26 (Leonardo's theorem). Every finite group of orthogonal transformations of the plane is cyclic or dihedral — where, to make the statement true as written, the dihedral family is extended downwards to D_1 (identity and one reflection, order 2) and D_2 (two perpendicular reflections and the half-turn: the Klein four-group of order 4), which Definition 1.9 does not name because no n -gon with $n < 3$ realises them. (The result is traditionally attributed to Leonardo da Vinci, who catalogued the possible symmetries of buildings.) We will not need the proof, and nothing later in the course depends on this remark: it is here for perspective, and may be skipped. The analogous three-dimensional classification, which governs the possible shapes of molecules and underlies crystallography, is taken up in Lecture 2.

1.5 Symmetry groups in physics: point groups

Model a molecule by the equilibrium positions of its nuclei — a finite set of labelled points in space, the labels recording the chemical species. The *point group* of the molecule is the group of all orthogonal transformations about its centre of mass (rotations, reflections, and their combinations) that map the configuration onto itself, carrying each nucleus to the position of a nucleus of the same species. “Point group” records that, unlike the translational symmetries of a crystal, all these operations fix a point.

Chemists’ (Schoenflies) notation will be useful: E for the identity, C_n for a rotation through $2\pi/n$ — the same symbol as the cyclic group it generates — and σ for a reflection plane, with a subscript v (*vertical*) when the plane contains the *principal* rotation axis, the axis of highest order; distinct vertical planes are told apart by primes, σ_v, σ'_v . A group name such as C_{2v} lists the principal axis (C_2) and the kind of reflection planes present (v).

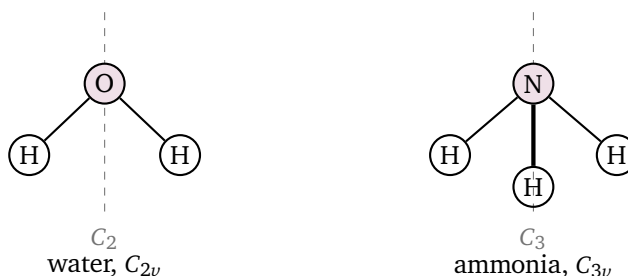


Figure 1.2: Two molecules and their point groups. Water is planar; its C_2 axis lies in the molecular plane. Ammonia is a shallow pyramid (the thick bond points towards the reader): the nitrogen sits above the plane of the three hydrogens, on the threefold axis.

Example 1.27 (Water). The water molecule H_2O is planar and isosceles (Figure 1.2, left). Its symmetry operations are: the identity E ; the rotation C_2 through π about the axis through the oxygen, bisecting the $\text{H}-\text{O}-\text{H}$ angle; the reflection σ_v in the molecular plane; and the reflection σ'_v in the plane perpendicular to the molecular plane containing the axis. The point group, denoted C_{2v} , therefore has order 4. Composing any two distinct nonidentity operations yields the third — for instance $\sigma_v \sigma'_v = C_2$, since reflecting in two perpendicular planes is a rotation through π about their line of intersection — and every element squares to the identity. The group is abelian. It is *not* cyclic (C_4 contains an element of order 4; here there is none), so $C_{2v} \not\cong C_4$: already at order 4 there is more than one group. This

four-element group, the *Klein four-group*, reappears on Sheet 1 as the symmetry group of a rectangle and of an anisotropic oscillator.

Remark 1.28 (The relabelling the definition suppresses). The definition asks that an operation carry each nucleus to the position of a nucleus of the same species — not that it carry each nucleus back to itself — and the difference is worth making explicit. The physical point is made in the first two paragraphs; the bookkeeping that follows them is not needed again until Lecture 6, and may be left until then. In classical mechanics the two hydrogens are distinguishable particles. The rotation C_2 therefore does not leave the labelled configuration alone: it puts H_1 where H_2 was. Read literally, the original labelling is restored only if the rotation is followed by the interchange of the two labels.

That interchange is not extra data. The nuclei sit at distinct points, so an operation g determines which nucleus lands on which, hence a permutation $\pi(g)$ of the nuclei of each species, defined by $g(\vec{p}_a) = \vec{p}_{\pi(g)a}$ when $\vec{p}_a$ is the position of nucleus a . Composing, $(gh)(\vec{p}_a) = g(\vec{p}_{\pi(h)a}) = \vec{p}_{\pi(g)\pi(h)a}$, so $\pi(gh) = \pi(g)\pi(h)$: the assignment $g \mapsto \pi(g)$ preserves multiplication, sending the point group into a symmetric group. (Lecture 2 names such a multiplication-preserving map a *homomorphism*.) For water it is the map into S_2 given by

$$E \mapsto e, \quad C_2 \mapsto (H_1 H_2), \quad \sigma_v \mapsto e, \quad \sigma'_v \mapsto (H_1 H_2).$$

It is not injective. The reflection σ_v is in the molecular plane, which contains all three nuclei, so it moves no nucleus at all although it is not the identity of $O(3)$: both E and σ_v are sent to the identity permutation. An operation and the permutation it induces are genuinely different objects, and passing from one to the other can lose information.

What licenses working with the bare operation is physics rather than convention. Nuclei of the same species are dynamically identical, so the Hamiltonian is left unchanged both by the orthogonal map and by any permutation of like nuclei. “Rotate, then relabel” and “rotate” therefore have the same effect on every physical quantity, and it is the rotation — an element of $O(3)$, fixed independently of how the nuclei happened to be numbered — that we take as the group element.

The permutation is not discarded. From Lecture 6 onwards it does real work: the number of nuclei that g leaves in place is the number of fixed points of $\pi(g)$, and that count is what turns a molecule’s geometry into the character of its displacement representation, and so into its vibrational spectrum.

Example 1.29 (Ammonia). The ammonia molecule NH_3 is a triangular pyramid: three hydrogens at the vertices of an equilateral triangle and the nitrogen on the axis above its centre (Figure 1.2, right). The symmetry operations are the identity, the rotations C_3, C_3^2 about the vertical axis, and three reflections in the vertical planes each containing the axis and one hydrogen. Note what is *absent*: the reflection in the plane of the hydrogens is not a symmetry (it would move the nitrogen below the plane), and likewise the π -rotations that fix an equilateral triangle. The point group, denoted C_{3v} , has order 6, and sending $C_3 \mapsto r$, $\sigma_v \mapsto s$ matches its multiplication with that of D_3 :

$$C_{3v} \cong D_3 \cong S_3.$$

One abstract group, three guises: the rigid motions of a triangle, the permutations of three objects, and the symmetry of a real molecule.

These examples illustrate the general procedure: enumerate the operations, close them under composition, and identify the resulting abstract group. What the group is *for* — how

it organises the molecule's vibrational spectrum, and in what precise sense a group is a symmetry of the *dynamics* rather than merely of a shape — is the business of Lectures 6 and 7, once the representation theory of Lectures 3–5 is in hand.

1.6 Subgroups

Symmetry groups come with natural subcollections: the rotations inside D_n , the even permutations inside S_n . These are groups in their own right, sitting inside a larger one.

Definition 1.30 (Subgroup). A subset $H \subseteq G$ of a group is a *subgroup*, written $H \leq G$, if H is itself a group under the multiplication of G .

Proposition 1.31 (Subgroup test). A subset $H \subseteq G$ is a subgroup if and only if

- (i) $H \neq \emptyset$;
- (ii) $h, k \in H$ implies $hk \in H$ (closure);
- (iii) $h \in H$ implies $h^{-1} \in H$.

If G is finite, conditions (i) and (ii) alone suffice.

Proof. Necessity of (i)–(iii) is clear (note $e_H = e$: the identity of H satisfies $e_H e_H = e_H$, and cancellation in G gives $e_H = e$). Conversely, given (i)–(iii): associativity is inherited from G ; picking any $h \in H$, closure and (iii) give $e = hh^{-1} \in H$; so H is a group.

Now let G be finite and assume only (i) and (ii). For $h \in H$, closure puts all powers $h, h^2, h^3, \dots$ in H ; by Proposition 1.7, h has some finite order n , so $h^n = e \in H$ and $h^{n-1} = h^{-1} \in H$, establishing (iii). (For infinite G the finiteness is essential: $\mathbb{N} \subset \mathbb{Z}$ is closed under addition but is not a subgroup.) ■

Example 1.32. (i) Every group has the *trivial* subgroup $\{e\}$ and the whole group G itself; subgroups other than G are called *proper*. (ii) The n rotations $\{e, r, \dots, r^{n-1}\}$ in D_n form a subgroup isomorphic to C_n : products and inverses of rotations are rotations. (iii) For any $g \in G$, the set of all powers $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$ is a subgroup, the *cyclic subgroup generated by g* ; if g has finite order then $|\langle g \rangle| = \text{ord}(g)$, since the distinct powers are exactly $e, g, \dots, g^{\text{ord}(g)-1}$. (iv) The even permutations form a subgroup $A_n \leq S_n$, the *alternating group*: by multiplicativity of sgn (Proposition 1.19), products of even permutations are even, and the finite subgroup test applies. For $n \geq 2$ exactly half of all permutations are even, $|A_n| = n!/2$: fixing a transposition τ , the map $\sigma \mapsto \tau\sigma$ is a bijection from even permutations to odd ones. The group A_4 , of order 12, will be a recurring character; it is the rotation group of the regular tetrahedron.

1.7 Cosets and Lagrange's theorem

How large can a subgroup of a finite group be? The answer is strikingly rigid, and the mechanism of the proof — slicing a group into translates of a subgroup — is used constantly in everything that follows.

Definition 1.33 (Coset). Let $H \leq G$ and $g \in G$. The *left coset* of H by g is the set

$$gH = \{gh : h \in H\} \subseteq G.$$

(Right cosets Hg are defined analogously; in Lecture 2 the comparison of the two will lead to the notion of a normal subgroup.)

Example 1.34. In $G = D_3$ take $H = \langle s \rangle = \{e, s\}$. The left cosets are

$$eH = \{e, s\}, \quad rH = \{r, rs\}, \quad r^2H = \{r^2, r^2s\},$$

and no others: e.g. $sH = \{s, e\} = eH$ and $(rs)H = \{rs, r\} = rH$. The three cosets are disjoint, each has two elements, and together they exhaust the six elements of D_3 . None of this is an accident.

Lemma 1.35. Let $H \leq G$. Then:

- (i) $gH = g'H$ if and only if $g^{-1}g' \in H$;
- (ii) any two left cosets are either equal or disjoint, and the cosets partition G ;
- (iii) every coset has exactly $|H|$ elements (when H is finite).

Proof. (i) If $g^{-1}g' = h_0 \in H$, then $g'H = gh_0H = gH$, since $h_0H = H$: the map $h \mapsto h_0h$ sends H into H by closure, is injective by cancellation, and is onto H because $h \mapsto h_0^{-1}h$ is an inverse for it. Conversely if $gH = g'H$ then $g' = g'e \in g'H = gH$, so $g' = gh$ for some $h \in H$, i.e. $g^{-1}g' = h \in H$.

(ii) Suppose $gH \cap g'H \neq \emptyset$, say $gh = g'h'$. Then $g^{-1}g' = h(h')^{-1} \in H$, so $gH = g'H$ by (i). Since $g = ge \in gH$, every element lies in some coset; hence the distinct cosets partition G .

(iii) The map $H \rightarrow gH, h \mapsto gh$, is surjective by definition and injective by cancellation. ■

Theorem 1.36 (Lagrange). If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$. More precisely, if $[G : H]$ denotes the number of distinct left cosets of H in G — the index of H — then

$$|G| = [G : H] \cdot |H|.$$

Proof. By Lemma 1.35, G is partitioned into $[G : H]$ cosets, each of size exactly $|H|$. ■

Corollary 1.37. Let G be a finite group. Then:

- (i) $\text{ord}(g)$ divides $|G|$ for every $g \in G$;
- (ii) $g^{|G|} = e$ for every $g \in G$;
- (iii) if $|G| = p$ is prime, then $G \cong C_p$; in particular G is cyclic and abelian.

Proof. (i) $\text{ord}(g) = |\langle g \rangle|$, which divides $|G|$ by Lagrange. (ii) Write $|G| = \text{ord}(g)m$; then $g^{|G|} = (g^{\text{ord}(g)})^m = e$. (iii) Pick $g \neq e$. Then $|\langle g \rangle| > 1$ divides p , hence equals p , so $G = \langle g \rangle$ is cyclic of order p ; by Example 1.21 it is isomorphic to C_p . ■

Part (ii) is the group-theoretic fact behind Fermat's little theorem and Euler's theorem in number theory. For us, Lagrange's theorem is above all a constraint on the search: it dramatically restricts what subgroups and element orders are possible. Lecture 4 will add a counting constraint of the same flavour on the dimensions n_i of the irreducible representations, $\sum_i n_i^2 = |G|$.

Example 1.38 (All subgroups of D_3). A subgroup of D_3 has order 1, 2, 3 or 6. Order 1: only $\{e\}$. Order 2: generated by an element of order 2 (Corollary 1.37(iii) with $p = 2$), i.e. by a reflection, giving $\langle s \rangle, \langle rs \rangle, \langle r^2s \rangle$. Order 3: cyclic, generated by an element of order 3, and the only such elements are r, r^2 , which generate the same subgroup $\langle r \rangle = \{e, r, r^2\}$. Order 6: D_3 itself. Exactly six subgroups. (You will carry out the same census for D_4 , where the answer is richer, on Sheet 1.)

Remark 1.39. The converse of Lagrange’s theorem is false: a divisor of $|G|$ need not be the order of any subgroup. The smallest counterexample is the alternating group A_4 , of order 12, which has no subgroup of order 6 — a fact we state without proof, since checking it needs more than the tools of this lecture. Partial converses exist — notably Cauchy’s theorem, that a prime divisor of $|G|$ is always the order of some element — and the case $p = 2$ appears as a guided exercise on Sheet 1.

1.8 Outlook: the landscape of finite groups

How many finite groups are there? Up to isomorphism — which, as Remark 1.23 explained, is the only sensible way to count — the numbers begin as follows.

order n	1	2	3	4	5	6	7	8	9	10	11	12
# groups	1	1	1	2	1	2	1	5	2	2	1	5

The entries 1 at the primes are Corollary 1.37(iii). At order 4 we have met both groups: C_4 and the Klein four-group of Example 1.27. At order 6 they are C_6 and $S_3 \cong D_3 \cong C_{3v}$ — the smallest nonabelian group, and the one this lecture has examined from three directions. The count is wildly irregular: it explodes at powers of 2 (there are 14 groups of order 16, and most groups of order at most 2000 have order exactly 1024).

A deeper organising principle, developed in Lecture 2, is that finite groups are built from *simple* groups — those admitting no nontrivial quotients — somewhat as integers are built from primes and molecules from atoms. The classification of the finite simple groups, completed in the late twentieth century after tens of thousands of journal pages, is among the largest collaborative achievements in mathematics; its strangest entry, the Monster group, with about 8×10^{53} elements, has unexpected connections to string theory. None of this machinery is needed in this course, but it is well to know the shape of the country whose border region we now begin to survey: in Lecture 2 we study how groups map to one another and decompose, and from Lecture 3 the central theme becomes how groups *act on linear spaces* — which is precisely the form in which symmetry enters quantum mechanics.

Remark 1.40 (Computer algebra). Mathematica handles finite groups natively, representing them as groups of permutations. Nothing is lost by this: Cayley’s theorem, which we do not prove, says that every finite group is isomorphic to a group of permutations, and Example 1.22 is one instance. Thus `DihedralGroup[3]` and `SymmetricGroup[3]` construct D_3 and S_3 ; `GroupOrder` and `GroupElements` return the order and the element list; `GroupMultiplicationTable` reproduces the Cayley table of Example 1.11, as indices into the element list; permutations are composed with `PermutationProduct`. A short notebook accompanying this lecture verifies the D_3 table and the subgroup census of Example 1.38. Beware one convention trap: Mathematica multiplies permutations *left to right*, opposite to these notes.

Summary.

- *Group.* A set with an associative product, an identity, and inverses (Definition 1.1); abelian when the product commutes.
- *Standard families.* The cyclic groups, the dihedral group D_n of order $2n$ (symmetries of the n -gon, Proposition 1.10), and the symmetric group S_n with its multiplicative sign map (a *homomorphism*, in Lecture 2’s language) (Definition 1.18).

- *Isomorphism*. A structure-preserving bijection; groups are studied up to isomorphism (Definition 1.20).
- *Subgroups and cosets*. The subgroup test (Proposition 1.31); cosets partition a group into blocks of equal size (Lemma 1.35).
- *Lagrange*. $|H|$ divides $|G|$ for $H \leq G$ (Theorem 1.36); hence element orders divide $|G|$ and groups of prime order are cyclic (Corollary 1.37).

Tutorial. Exercise Sheet 1.

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